quadrant · per-slide label sizing

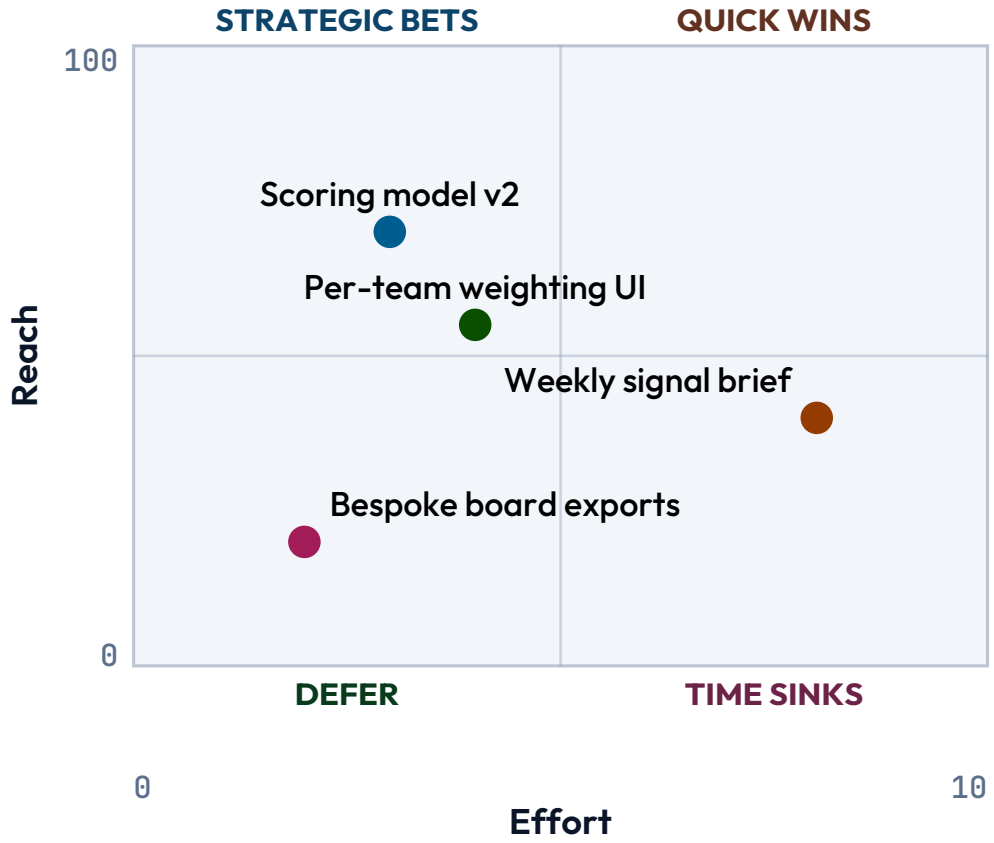
One crowded slide set the type for all of them.

A quadrant prints each item's name beside its dot. That name had **one size**, written once in the kernel and shared by every quadrant in every deck — and the number had to be the largest the most crowded slide in the corpus could carry. So a four-dot chart with a half-empty plot printed at the size a fourteen-dot chart needed, in a different file, written by someone else.

The constant was a floor disguised as a size.

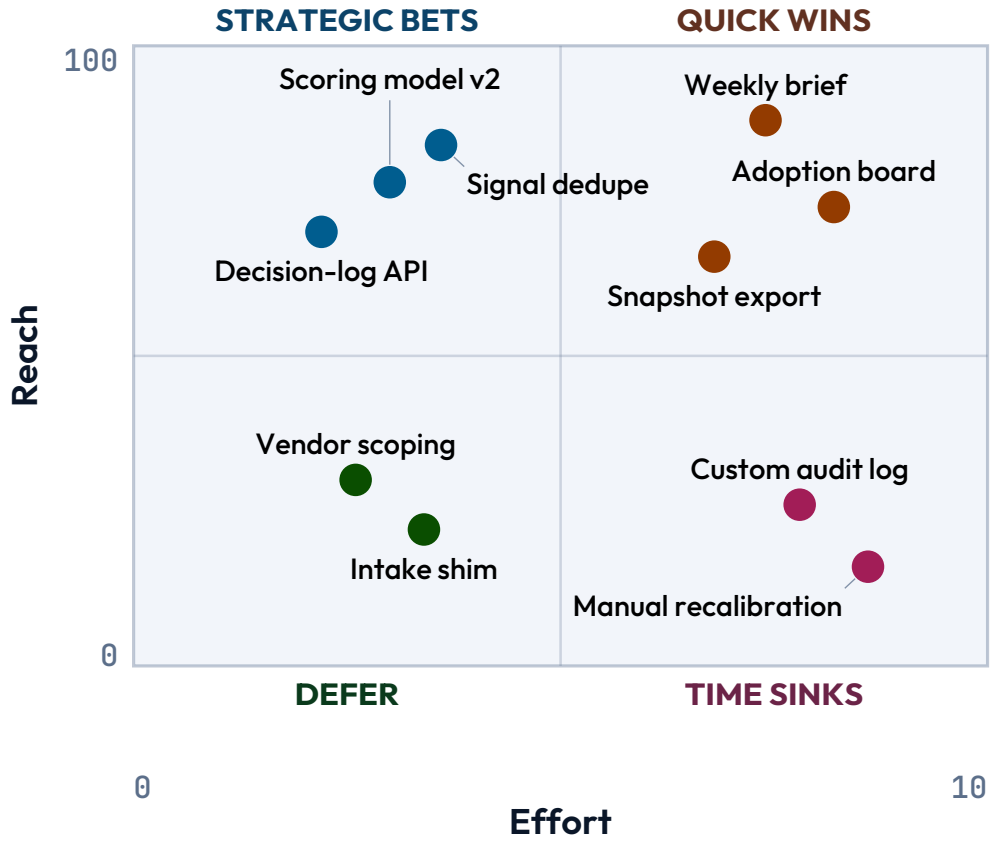
`FS.dotLabel` sat at **9.5** because the fourteen-initiative stress slide loses a name at 10.5 — which is exactly what happened once, in #1544, and it reached a committed gallery PDF before anyone caught it. The number was correct for that slide and wrong for the twenty-five others paying for it. Measured across the corpus: **twenty-five of twenty-seven** quadrant slides were running below what their own geometry allows, and not one reached the house's smallest type tier.

Where to put the next dollar.



Four items, a half-empty plot — now sized to the room it has

Ten initiatives, and every name still reads.



Ten items — still clears the bar, because names are short

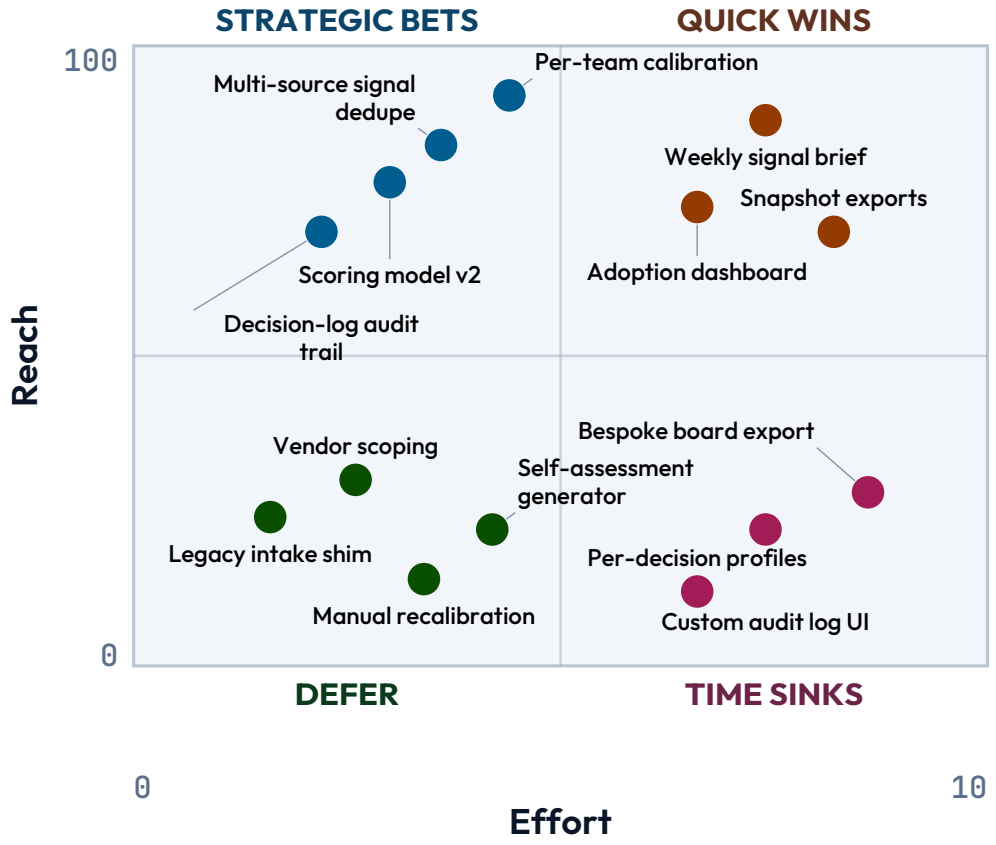
The size is asked of the geometry, not chosen.

The kernel walks a ladder of sizes **largest first** and keeps the first rung where every name still places. `placeLabels` is pure and deterministic, so this is a measurement, not a heuristic. The bottom rung is the old constant — a slide the ladder cannot lift renders **byte-identically** to before.

The box grows with the type, and that was not optional.

`LW.item` is a width in viewBox units. It buys *characters* only in combination with a font size — held at 120 while the type grew, a line fell from about twenty-one characters to fifteen, and names began breaking mid-word: `Maturity self-assessment` rendered as **self-assessmen / t**. Scaling the budget by the same factor holds characters-per-line constant, so a label is a larger copy of itself.

The density that pins the floor.



Fourteen items — the ceiling, and unchanged at the floor size

The ceiling is a design decision, not a limit we hit.

Plenty of sparse quadrants place every name at 18 or 20, and they are **not allowed to**. A dot name that outgrows `--quadrant-axis-size` inverts the chart's hierarchy — the thing being measured would print larger than the axis it is measured against. The ladder stops at the house's smallest type tier: reaching the legibility bar is the goal, and passing it is a different defect.

What this does not fix.

Six slides still cannot carry a readable name, and count is only half of why — name **length** and dot proximity bind too, so some six-item slides settle a rung below every ten-item slide. Those keep the floor size: no worse, no better. Warning an author *before* they build a fourteen-dot quadrant needs a signal the chart family does not have; it is tracked separately.